

SAI SAMHITHA MUDIGONDA

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PROFESSIONAL SUMMARY

Full-Stack Engineer with experience in building large-scale, customer-centric applications and optimizing distributed systems within high-traffic cloud environments. Specialist in Micro-frontend SPA architecture and BFF (Backend-for-Frontend) design patterns, focused on delivering seamless and high-performance user experiences. Proficient in a modern technical stack including React, Next.js, Node.js, TypeScript, and Java (Spring Boot) to build scalable and maintainable software solutions. Proven track record of enhancing system reliability and CI/CD efficiency by implementing AI-driven automation and advanced monitoring through Amazon Web Services. Strong communicator skilled at translating complex business requirements into technical roadmaps while providing mentorship to guide junior engineers in technical growth.

PROFESSIONAL SKILLS

Languages: JavaScript, Next.js, TypeScript, Java (Spring Boot), Kotlin, Python, SQL, HTML5, CSS3

Architecture: Micro-frontend SPA, BFF (Backend-for-Frontend), Microservices, API Gateway, Distributed Systems

Frontend: React.js, Redux, GraphQL, tRPC, Responsive Design, Accessibility (WCAG), Component Architecture

Backend: Node.js, REST APIs, GraphQL, Spring, PostgreSQL, CloudFormation, API Integrations

Cloud & DevOps: Amazon ECS, Azure, GCP, Kubernetes (EKS), Lambda, CI/CD (Jenkins), CDK, CloudWatch, S3, DynamoDB

Testing & Tools: AI-driven development tools, Jest, Cypress, PyTest, Git, Webpack, Agile/Scrum

WORK EXPERIENCE

Amazon Web Services (AWS)

Software Development Engineer

Jersey City, NJ

April 2025 – Present

- Managed the development and maintenance of the AWS ECS Console using React, TypeScript, and Node.js to facilitate large-scale container orchestration for millions of users.
- Architected modular Micro-frontend components and SPA structures that enabled the Experience Development Team to deliver independent features without disrupting the core system.
- Implemented BFF (Backend-for-Frontend) design patterns to act as a translation layer between complex frontend requirements and distributed backend microservices for better efficiency.
- Resolved critical root causes within distributed service failures and API defects by performing deep-dive analysis, resulting in a significant rise in overall console uptime and reliability.
- Optimized PostgreSQL query performance and API Gateway configurations which reduced execution latency and provided a more responsive experience for container management tasks.
- Created a comprehensive regression testing framework utilizing Python and AI-driven scripts to automate manual QA efforts and accelerate the speed of bi-weekly release cycles.
- Refined CI/CD infrastructure and Jenkins pipeline configurations to increase the frequency of deployments while reducing the time required for system rollbacks during production issues.

Cognizant Technology Solutions (CTS)

Full Stack Developer Intern

Hyderabad, India

February 2022 – July 2022

- Constructed responsive web interfaces and enterprise application modules using HTML, CSS, and JavaScript to ensure consistent performance across various mobile and desktop devices.
- Developed reusable and modular UI components that followed Micro-frontend design principles to improve code maintainability and ensure visual consistency across the platform.
- Integrated frontend application modules with Node.js and Java backends to enable dynamic data rendering, secure form submissions, and real-time state updates for users.
- Debugged and fixed production frontend defects involving DOM rendering issues, asynchronous API call timing, and cross-browser compatibility to provide a stable user experience.
- Collaborated with QA engineers and backend developers within an Agile environment to deliver technical requirements and feature updates according to the project roadmap.
- Translated business functional requirements into technical specifications while participating in sprint planning to prioritize high-impact tasks for the development team.
- Provided technical support and documentation for newly developed features to assist team members in understanding the integration points between the frontend and backend services.

EDUCATION

University of Central Missouri
Master of Computer Science

Warrensburg, Missouri

CERTIFICATION

Google Cloud Facilitator Program – Google Cloud
Python Programming Certification – GUVI

Credential

Credential